Point decomposition beyond summation polynomials: literature review and 19 leads (two rounds)

Date. 2026-10-22 (dossiers completed 2026-10-23). **Type.** Literature search round with lead dossiers (research-visuals skill, "Deliverables for each search round"); no experiment was run. **Status of every statement.** *Proposed* unless a sentence says *derived* (desk algebra in a dossier, labelled unverified), *read at source* (a cited text was fetched and read), or *measured* (a number quoted from a frozen repository artefact, with its path). Nothing here changes a leaderboard, benchmark or scoreboard row, and no claim about a deployed curve is made or withdrawn. **Conductor.** Not installed in this container (the hooks warn and let edits through); the study touches only this directory.

Question. Which reformulations of the point decomposition problem (PDP) could replace or relax Semaev's summation polynomials in the relation search of elliptic-curve index calculus and lower the Gröbner-basis cost or the degree of regularity, measured against the current index-calculus baseline (`docs/ic/LEADERBOARD.md`: every row above matched rho, Koblitz 5.23×–21×, prime 5.47×–13.5×)?

Answer in one paragraph. The literature since 2020 is dormant on the summation-polynomial side (no new decomposition method; one 2026 preprint on the degree of regularity of descended S_3; learned Gröbner solvers and Crossbred theory on the solver side; two new generic-group lower bounds). Of the 14 leads drafted here, the director-tier dossiers close most of them: the MinRank, chord-lattice, trace-zero and prime-field block-structure reformulations are the same ideals or the same oracles the repository already has (LEAD-01, 06, 10, 14, each *closed-by-theorem* at the desk); the class filter and learned orderings are constant-factor levers that move no exponent and at most 1.1–2× at stage level (LEAD-04, 08); the McGuire–Mueller text and the elliptic-curve theorem of ePrint 2026/384 could not be read from this container (LEAD-11, 12, *unverifiable-from-here*); and the 2026 degree law is plausible for two-point S_3 but does not transfer to the chain routes (LEAD-05). The gate dossier (LEAD-12) finds no lead that the generic bounds exclude outright and three that could move the exponent in principle, which means "not excluded", not "indicated". The five remaining dossiers close their cards at the desk as well: the parametric eliminant needs about D^{k+1} terms against a $D^{3.1}$ solve and costs more to evaluate than the solve it replaces (LEAD-02); ℓ-torsion beyond 2 multiplies the solution count by more than the symmetry group can divide it, and the 3-torsion frame was already measured not to pay (LEAD-03); every admissible Crossbred choice on the deciding $(m, \ell) = (4, 6)$ cell enumerates more variables than the summands have (LEAD-07); the Ore resultant is dense of degree $2^{\ell+1}$ and its roots are the list rf-mitm already enumerates (LEAD-09); and the summation-ideal solution-count drop needs characteristic 2 while both repository cover paths are odd-characteristic (LEAD-13). The cheapest next measurement that decides several cards at once is LEAD-05's S_3 boundary replication (about 1 person-day, 10–15 CPU-hours), which LEAD-12 routes first; the only candidate for a measured improvement over a leaderboard row is LEAD-04's cofactor-projected base on the random-binary ladder (desk estimate 1.4–1.6× at fixed base size).

Round 2 (2026-10-23). Five further cards (LEAD-15 to LEAD-19) follow the dossiers' own pointers; their verdicts are in §10. One is an exponent-candidate (LEAD-15, Diem's plane-quartic branch of the cubic-GHS transfer), one is a constant-factor candidate with a desk estimate of 1.4–3× on three leaderboard rows (LEAD-19), and the rest close at the desk.

Follow-up (2026-10-23, same PR). LEAD-19's discriminating test was run under a frozen protocol: `../lead19_projected_base/`. The measured lever S(A′)/S(B) at the published rung is 1.29 (95 % interval 1.12–1.52), 1.58 (1.35–1.82) and 3.36 (2.80–4.02) on the three rows (desk estimates 1.4–1.5, 1.4–1.5, about 3), classified engineering; the desk text of this study, including §10, is left as written.

Figures: `figures/pdp_landscape.svg` (source `figures/diagram.py`) and `figures/dreg_records.svg` (source `figures/plot_dreg.py`, data `figures/dreg_records.csv`). PDF: `report.pdf` (source `report.typ`). Literature map: `papers.csv`. Cards: `LEADS.md`. Dossiers and verification cards: `leads/`.

1. Scope and method

1. **Repository sweep** (2026-10-22, one read-only agent over `crypto`, `cryptanalysis`, `crypto-autoresearcher`, `ml-cryptanalysis`, `cryptanalysis-rl`): existing literature notes, code modules, ledger records, open problems, negative results and the baseline. The programme already holds a 25-file known-results map and about 60 literature notes on this topic in the autoresearcher's `knowledge/`, the master boundary ledger, two syntheses (`reports/Index calculus leads next steps.md`, `research/ic_leads_review_20261018/`) and native oracles for Semaev, Nagao, SAT/WDSat, root finding and Crossbred. The cards were written against that map so that closed ground is not re-opened (list in `LEADS.md`, "Standing constraints").
2. **External literature** (about 30 web searches and 12 page fetches, 2026-10-22): the summation-polynomial line from Semaev 2004 to Galbraith–Granger–Merz–Petit 2020, the degree-of-regularity line from Petit–Quisquater 2012 to Kousidis–Wiemers 2019 and the 2026 Solovyev preprint, the Gröbner-free and alternative-formulation line (Nagao, Faugère–Wallet summation ideals, McGuire–Mueller, Di Tullio–Pal, Gorla–Massierer trace zero), the solver line (WDSat, Crossbred, learned Gröbner bases) and the bound line (structured generic-group model, non-adaptive time–space bounds). `papers.csv` lists 44 sources with the access status each one had from this container: IACR ePrint PDFs are Cloudflare-blocked here, so most ePrint items are abstract-only.

3. **Lead cards** (`LEADS.md`): 14 cards, each with mechanism, why it could lower the Gröbner cost or the degree of regularity, prior art, the closed ground nearby, a discriminating test with a kill criterion, and repository hooks.
4. **Dispatch** (owner's model tiers, 2026-10-17): one director-tier Sonnet 5.5 agent per card wrote `leads/LEAD-NN.md` under `BRIEF_SONNET.md` (literature first, every citation marked read-at-source / abstract-only / recalled, no experiment, no research-state change); one mechanical-tier Haiku 5.5 agent per dossier wrote `leads/LEAD-NN.verify.md` under `BRIEF_HAIKU.md` (every citation re-fetched, every path tested with `test -e`, every quoted number grepped; computes nothing). The container restarted once during the round; the five dossiers that were lost (02, 03, 07, 09, 13) were re-dispatched on 2026-10-23 with a time budget. The second LEAD-07 agent stalled for three hours and was stopped; a third, with a 25-minute budget, wrote the dossier from the repository files, the published version of ePrint 2024/992 (read at source) and the two other ePrint abstracts.

2. What the literature says (summary of `papers.csv`)

- **The formulation is unchanged since 2004.** Every index calculus on elliptic curves over extension fields decomposes `Q = P₁ + … + P_m` by the zero set of Semaev's `S_{m+1}` (degree `2^{m−1}` in each variable), Weil descent to `n` equations in `m·ℓ` unknowns, symmetrisation under `S_m` (Faugère–Perret–Petit–Renault 2012) and under rational 2-torsion (Faugère–Huot–Joux–Renault–Vitse 2014), and a Gröbner or XL solve. The two published alternatives to `S_{m+1}` are Nagao's Riemann–Roch system (`(n²−n)g` quadrics; 2007/2010) and Faugère–Wallet's summation ideals on hyperelliptic Jacobians (2018), both for Jacobians on covers; the repository measured Nagao at prime `n` as "`S′₄` with the linear part hidden" (`RESEARCH_PDP_SAT_SMT_NAGAO_SURVEY.md` §5). McGuire–Mueller (ePrint 2017/1262) claim a Gröbner-free evaluation algorithm and a second one without summation polynomials; the text could not be read here (LEAD-11). Di Tullio–Pal (arXiv 1909.08901, read at source) write the prime-field DLP as quadratic conditions on functions `y·g(x) + h(x)` with a prescribed divisor, which is the Riemann–Roch form again.
- **The degree of regularity is the open quantity, and the literature is split on what to call it.** Petit–Quisquater's first fall `m²+1`, Kousidis–Wiemers' `m²−m+1`, Semaev's "`≤ 4` under the first-fall-degree assumption", Huang–Kosters–Yeo's evidence against that assumption, and Kosters–Yeo's first fall 2 for descended `S₃` all coexist. The only recent data point is Solovyev's 2026 preprint (GitHub release, no venue, single author): exact degree-capped closures giving `S₃` degree 3, 4, 5 at `n ≤ 16`, `17–36`, `37–52`, then linear growth, with closed-form closure dimensions and `S₄` solving degree = first fall = 7 up to `N = 24`. Figure 2 plots the recorded values side by side with the repository's own msolve (4) and XL (5–7) readings on the same cells.
- **Solvers moved, formulations did not.** WDSat (Trimoska–Ionica–Dequen 2019–2021) is measured here as an enumerator (`2^{3ℓ}/3!` nodes to within 0.2 %); Crossbred theory (Vidal–Ionica–Delaplace, ePrint 2024/992; preprocessing ePrint 2026/360) was never tried on the descent cells; learned Gröbner bases (Kera et al., NeurIPS 2024; HATSolver, arXiv 2512.14722) predict bases for 0-dimensional radical ideals and do not learn orderings (the LEAD-08 dossier corrects the card on this point).
- **Two 2025–2026 lower bounds fence the oracle space.** The structured generic-group model (Corrigan-Gibbs–Henzinger–Wu, ePrint 2026/384) and the non-adaptive time–space bounds (Dinur–Keller–Marmor, ePrint 2025/783) make every table-lookup or enumeration oracle a constant-factor lever; LEAD-12 maps each card onto them and records which statement is the paper's theorem and which is this repository's interpretation (the elliptic-curve theorem of 2026/384 was not readable from here).

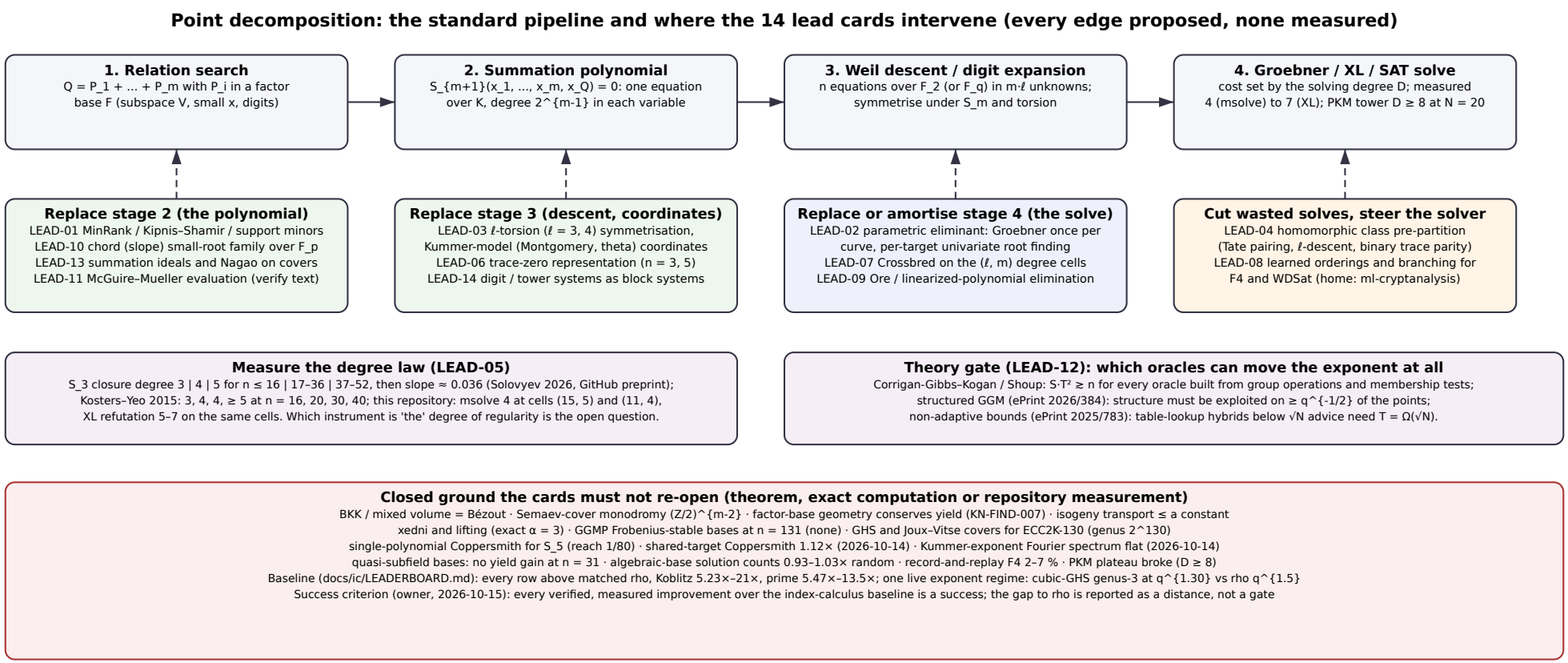


Figure 1. The standard PDP pipeline and where each lead card intervenes. Every edge is proposed; none is measured. Source: figures/diagram.py.

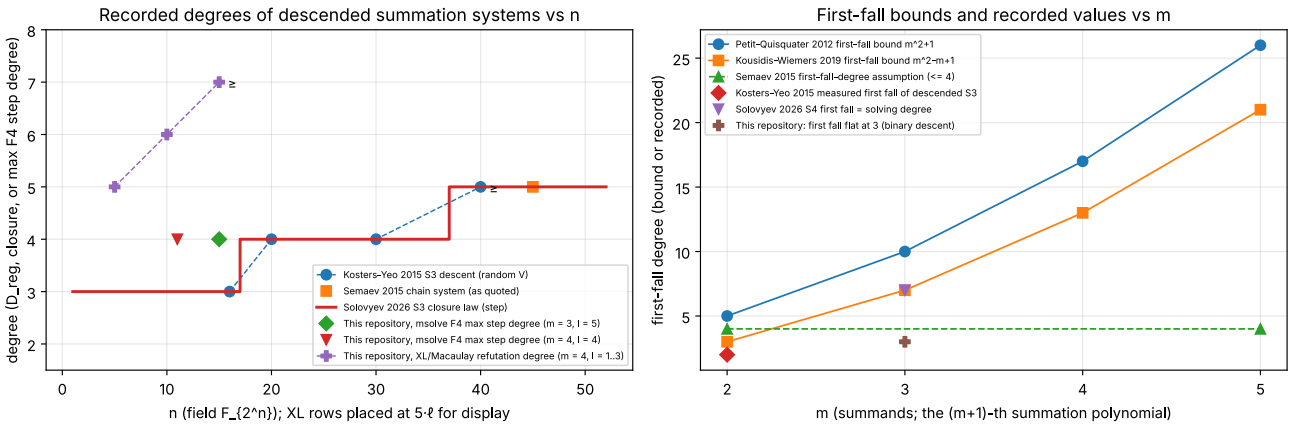


Figure 2. Recorded degree values for descended summation systems (left: degree versus n; right: first-fall bounds and recorded values versus m). Literature values are quoted as recorded, repository values name their file in figures/dreg_records.csv; no new computation. Source: figures/plot_dreg.py.

3. The 14 leads: verdicts

Verdicts are the dossiers' own (`leads/LEAD-NN.md`); "verification" is the mechanical card's totals (`leads/LEAD-NN.verify.md`: citations verified / mismatch / unreachable; paths exist / missing). Every mismatch and missing row was read by hand and is classified in §5; none is a wrong citation.

Lead	Card (class)	Dossier verdict
01	MinRank / Kipnis–Shamir / support-minors modelling of the summation chain (formulation)	**closed-by-theorem**: the Minors, KS and SM modellings are the repository's S_4-direct system, the chain, and the chain's ideal; none can read a lower degree than the chain. One residual probe (carry seeds) keeps about a 10 % prior of one degree
02	Parametric eliminant: generic-fibre Gröbner basis in the target's coordinates (solver, amortisation)	**closed-by-measurement**, provisional (desk): over $\bar{F}_q$ the descended $k = 3$ system splits into one equation per conjugate target coordinate; the solution count is 384 before and 64 after S_3-symmetrisation, so the eliminant has degree 64 in e_1 and up to 64 in each of the three target coordinates, up to $65^4 \approx 1.8 \cdot 10^7$ terms and at least 274,625 specialisations; evaluating it per target costs 4–60× the measured C_3 at $k = 3$ and about $9 \cdot 10^4 \times C_4(D_4)$ at $k = 4$, because an eliminant in k target coordinates needs about D^{k+1} terms against a $D^{3.1}$ Macaulay solve
03	ℓ-torsion (3, 4) symmetrisation and Kummer-model coordinates (formulation)	**closed-by-measurement** for $\ell = 3$ (the 3-torsion frame was measured at $m = 3$ in `RESEARCH_EXOTIC_COORDINATES.md` §14.3, "does not pay"), **closed-by-theorem** for $\ell = 4$ and for elliptic Kummer/theta-line coordinates (FHJRV Prop. 3 needs $\ell \mid \deg \phi$; Prop. 5 makes every Kummer-line coordinate a Möbius image of x): ℓ-torsion forces $\deg \phi \geq \ell$, breaks the $[-1]$ halving and raises the solution count from $2^{(m-1)^2}$ to ℓ^{m^2-m+1}, which the group order $\ell^{m-1}\, m!$ cannot recover; the Hessian S'_3 has degree 9 per variable against 2 (exact check), so the card's kill fires at $m = 2$
04	Homomorphic class pre-partition: Tate pairing, ℓ-descent, trace parity (filter)	**constant-factor candidate**, cofactor curves only: Z/r has no proper quotient, so every prime-order row (all eight Table B prime rows) gets exactly 1.000×; pigeonhole caps the $m = 3$ probe saving at 2× for any class group, so the card's "$\|A\|$ in every oracle" is wrong
05	Degree-of-regularity growth law for descended S_3/S_4 (measurement)	**unverifiable-from-here**: the S_3 thresholds $3 \longrightarrow 4 \longrightarrow 5$ at square $N = 16, 36$ agree with Kosters–Yeo, Kousidis–Wiemers and the repository's own `lfd.sage` boundary, with one conflicting cell (Magma reads 4 at $n = 35$ where the preprint reads 5); the law is for the quadratic two-point S_3, not the cubic-plus-quadratic chain, and the preprint's own S_4 data are flat at 7
06	Trace-zero variety representation for $E(F_{q^n})$, $n = 3, 5$ (representation)	**closed-by-theorem**: T_n exists only for a curve defined over F_q, so it is not a coordinate system for the baseline's random prime-order E/F_{p^k}; at $n = 3$ it ties rho at q^1; at $n = 5$ the T_5 system is 9 equations of degree 32 in 8 unknowns against the $k = 4$ ladder's 4 equations of degree ≤ 8

07	Crossbred on the Weil-descent degree cells (solver)	**constant-factor candidate**, lowest rung, with the card's own kill (`k > ℓ`) expected to fire at `(m, ℓ) = (4, 6)`: only `mℓ = 24` of the `N` Boolean variables are free, and every admissible Crossbred choice in the dossier's desk model (a semi-regular count extended from the paper's quadratic theorem to mixed quadratic and cubic equations) enumerates `k ≥ N − 8`, i.e. 38–64 variables against 24; with `k ≤ ℓ` the specialised system is under-determined at `n = 10` and 15 and no `D ≤ 14` is admissible at `n = 24`; preprocessing over all `N` variables is about 9× wider than guess-then-F4 on the `m = 3` residual; the card's "70 GB at degree 5" is the `N = 40` figure, and `(4, 6)` at `n = 10` is 191 GB dense
08	Learned orderings and branching for F4 and WDSat (solver, ML)	**constant-factor candidate**, stage level only: a variable order cannot change F4's solving degree on refuted or unique-solution targets (Caminata–Gorla Thm 3.1, Cor 3.3, read at source) and cannot lower WDSat's node count (thesis tables within +0.8 % of `2^{3ℓ}/3!`); only the F4 word-XOR cost is in play, where hand rules already give 2–4.91× at stage level with `S` unchanged
09	Ore / linearized-polynomial quotient-algebra elimination (solver)	**closed-by-measurement** (desk, exact over `F_{2^n}` at `ℓ = 3…8`) for the card as written: `Res_{x₂}(S₃, L_V)` has degree exactly `2^{ℓ+1}` in `x₁` (proved) and is dense (at most one zero coefficient in 16 instances, full ANF support), so the fast skew-polynomial costs, which are in the 2-degree `ℓ`, apply to `L_V` and its compositions (already used in `semaev_decomp.rs`) but not to the resultant; its roots are the candidate list `x(R ± P_v)` that rf-mitm already enumerates at 5 multiplications each, a membership-test oracle in disguise
10	Chord-parametrised small-root family over `F_p` (formulation)	**closed-by-theorem** for the small-unknown family: the chord system at `m = 2` is Semaev's `S₃` in the symmetric `(e₁, e₂)` encoding after a free linear elimination of the slope (symbolic and numeric desk check); every encoding has the same limit reach `2/(m(m+1)d)` = 1/6, 1/24, 1/80 and ceiling `1/(md)` = 1/4, 1/12, 1/32, so the `m = 4` kill fires (1/80 < 1/32 < 1/5)
11	McGuire–Mueller Gröbner-free algorithms, ePrint 2017/1262 (literature verification)	**unverifiable-from-here**: the PDF is Cloudflare-blocked (`cf-mitigated: challenge`), the Caen server times out, no mirror found; provisional classification from second-hand evidence: Algorithm 2 (sum `m−1` base points, test membership) coincides with the repository's `rf-pairs`/`rf-mitm` oracles, Algorithm 1 is the same oracle with an algebraic evaluation step; both on the CGK line
12	Theory gate: structured GGM and non-adaptive bounds (theory)	**unverifiable-from-here** for the one statement it most wants (the elliptic-curve theorem of ePrint 2026/384 could not be read); the 14-row classification is delivered: 3 leads can move the exponent in principle (01, 06, 10 conditional), 9 constant only, 0 excluded outright, 2 are not oracles; none of CGK, Shoup, DKM or SGGM constrains an oracle that reads coordinates, so "in principle" means "not excluded"
13	Summation ideals (Faugère–Wallet) and Nagao on Joux–Vitse covers (formulation on the cover)	**closed-by-theorem** for the repository's cover paths: the `2^{(n−1)(g+1)}` drop needs characteristic 2 and a monomial `h₁` (2-rank at most 1), and Wallet's 2016 slides say "no reduction in characteristic `p > 2`"; both repository cover paths are odd-characteristic, the cubic-GHS `q^{1.30}` route has no Gröbner step at all (smoothness tests with large primes), and in the Joux–Vitse route the Nagao test survives only in the descent (53–56 % of `S` at `p = 1511`); the frozen `experiments/30_jv_cover_ccov.json` shows ideal degree 64 in all 10 rows, so the card's kill (an 8× fall) fires; the Jacobian's rational 2-torsion gives no quotient for `g ≥ 2` because translation by a 2-torsion class is an automorphism of `J`, not of the cover curve
14	Prime-field digit and Kummer-tower systems as multi-homogeneous blocks (formulation)	**closed-by-theorem**: the bilinear and unmixed multilinear bounds do not apply (two structural hypotheses fail: the tower quadrics have block degree 2 and the system is over-determined); their formal values 7, 9, 11, 13 at `m = 3`, `N = 9–18` exceed the measured 6, 6, 7, 7 with a slope 3–7× too steep; any valid block bound has slope `(m−1)/m` in `N`, which the PKM note's §3.6 turns into at least `√r` per target

Table B. Ranking inputs and verification totals per lead.

Lead	Confidence	Expected gain if it works	Test cost	Verification (citations v/m/u; paths e/m)
01	medium	at most one degree (4 ⟶ 3): `m = 4` at `n = 131` moves +7.62 ⟶ +5.34 bits over rho (`w = 1`), still above	1–1.5 person-days, < 4 CPU-h, ≤ 1.5 GB (msolve)	12 / 2 / 0; 25 / 1

02	medium (degree counts desk-checked at `k = 3`; density measured only at `k = 2`, 67 of 125 terms)	none; if evaluation were free the ceiling would be 5–9× at `k = 3` and about 10^3× at `k = 4` (arithmetic, not a result); setup amortises at `k = 3` (about 9 % of the relation phase at `q = 2^20`) but not at `k = 4` before the extrapolated D_4 handover	2–3 person-days of Rust (a `k = 3` slice-fill measurement), < 1 CPU-h, < 1 GB	2 / 2 / 6; 8 / 0
03	high (counting argument, exact degree check and the repository's measured row agree)	none: at `m = 3` the descended Hessian system has 2,187 $\bar{F}_q$ solutions against 16 (2-torsion frame) and 64 (plain symmetrised Semaev); the measured F4 proxy is 5–8× slower than Gaudry's x-line; a monic Hessian with `ω ∈ F_q` has `9 | #E`, so the matched rho gets 3× cheaper (brute-force counts at `p = 7, 13, 19, 31`)	1 person-day for a Hessian chart in the quotient engine, < 5 CPU-min; recommended skipped	4 / 1 / 7; 11 / 0
04	medium (bounds); low (effect on `S`)	about 1.1–1.2× on `S` for the cofactor-2/6 random-binary rows (desk estimate); exponent unchanged	1 person-day, < 1 CPU-h	2 / 0 / 0; 30 / 0
05	medium (S_3 import); low (transfer)	none as an improvement; if the slope transferred, the `m = 3, 4` rows at `n = 131` would move *up*, to $2^{114.7-2}$129.1 and $2^{132.1-2}$155.6 column touches	1 person-day, 10–15 CPU-h, ≤ 12 GB	8 / 0 / 0; 24 / 1
06	high (no improvement); medium (`n = 5` asymptotics)	none against any baseline row; a `T_n` row would be a new baseline, not an improvement	2–3 person-days (new T_3 harness), ≤ 2 CPU-h	1 / 4 / 4; 16 / 0
07	low that it gains; medium that the kill fires (the desk model predicts degree 7–8 where the repository's F4 reads 4, so only the test can close it)	unknown, probably none: at most a constant on peak memory at `(4, 6)` and nothing on end-to-end `S`, because the `m = 4` free-oracle floor of `2^56.46` binds; the gap to rho stays at the committed +7.62 bits	3–5 person-days, 40–60 CPU-h at 16 GB; an optional degree-5 item adds about 15 CPU-h at 7–25 GB	3 / 2 / 1; 14 / 0
08	medium	0 degree levels; WDSat ≤ 1.04×; F4 XORs 1.0–2× beyond the hand rules (guess); `S` 1.00×	2 person-days core, 10 CPU-h mechanical, ≤ 16 GB	7 / 0 / 5; 21 / 3
09	medium (degree proved; density at ℓ = 14–16 extrapolated)	none: at `n = 47` writing out the resultant costs `2^{ℓ+1}+1` coefficients, about 0.80× rf-mitm's counted multiplications, at least 2.4× above the card's kill line; the `m = 3` pairs residual floor is 3.2–3.9× above rf-pairs; `S_IC/S_rho` unchanged (3,053 at `n = 31`, 25,116 at `n = 37`)	0.25 person-day, < 1 CPU-h, < 1 GB (a term-count example on `semaev_decomp::{Gf2, Poly}`)	6 / 2 / 1; 24 / 0
10	medium	none in the exponent; 5–11 % higher δ at equal multiplicity (2–3× smaller lattices) on oracles that cannot beat rho	zero-compute test done in the dossier; 0.5 person-day to confirm in the `05` MILP harness	6 / 0 / 1; 12 / 0
11	low (text); medium (classification)	none; the relayed `O(p)` cost is a worse exponent than Table B's `m = 2` row (`r^{2/3}`)	0.5 person-day once a host can download the CC BY PDF	9 / 2 / 3; 13 / 0
12	medium (generic rows); low (algebraic rows)	none (a gate); routes the first test to LEAD-05	0.5 person-day to close the reading gap	8 / 1 / 3; 10 / 0
13	high (repository path); medium (even-characteristic numbers: Faugère–Wallet's HAL and Springer pages were unreachable, so the abstract and Wallet's slides were used)	none on any row: 1.0× at `q ≈ 2^11` in odd characteristic; a hypothetical even-characteristic Joux–Vitse cover would have `L = g`, giving at best degree 64 ⟶ 32 (a symbolic recount of the norm polynomial finds 64 ⟶ 64), about 1–6× per test (desk estimate);	T0 is a read-only `jq` over the frozen JSON (0.1 person-day, already run: `delta.max = 64` in every row); the even-characteristic control T1 (3–5 person-days, external CAS) is not recommended	5 / 0 / 1; 19 / 0

		ECC2K-130 has no low-genus cover		
14	medium	none on the exponent, and the sign is wrong; constant unknown, likely ≤ 1.5×	0.5 person-day, < 1 CPU-minute	10 / 0 / 0; 12 / 1

3.1 What the completed dossiers add beyond their verdicts

- **LEAD-01.** The Sylvester determinant of the $S_3 \circ S_3$ chain is S_4 (desk check); Bardet–Bertin Prop. 1 (read at source) gives KS = SM ideals with Minors ⊂ KS, so the three modellings are generating sets of ideals the repository already solves. The bilinear bound does not apply because the chain has the cubic term x_1x_2y; read naively it gives 16 at `(15, 5)` against a measured F4 degree of 4. The free-kernel KS system is 14.7–16.4× wider at degree 4. The one thing left is whether the closed-form carry seeds (bit components of x_1f_1, x_2f_1, yf_1) lie in the degree-4 Macaulay span and let F4 finish at 3; the dossier replaces the card's test with that probe (R1).
- **LEAD-02.** Break-even at `k = 3` needs an eliminant fill of 1.6–2.5 %, which is the one unmeasured quantity (the test). Joux–Vitse's F4 traces were read at source from saved PDFs; Weispfenning 1992 is recalled, Montes–Wibmer and Faugère–Mou abstract-only.
- **LEAD-03.** The Hessian invariant ring is polynomial: `G(3,3,3)` has order 54 with primary invariants Σu_i^3, `Σ_{i<j}` $u_i^3u_j^3$ and $u_1u_2u_3$ and no secondary invariants, so the obstruction is the solution count, not the invariant theory. FHJRV stop at 2-torsion because degree-2 maps are automatically 2-torsion-equivariant. The Barbulescu–Robert–Sarkis PDF returned 503, so only its abstract was read.
- **LEAD-04.** Why KN-FIND-14706d does not cover this: that finding tests the product character $\chi_1\chi_2\chi_3$, which is trivial on `E(F_p)` by the curve equation; the 2-descent class (χ_1, χ_2) lives in its kernel and takes all four values. But on a prime-order row there are no classes at all, and on cofactor rows hash lookups and membership tests are already exact and uncharged, so only table-build additions fall. The dossier flags an adjacent lever outside the card: the random-binary ladder builds its base from raw points with no projection into the subgroup; a class-0 or cofactor-projected base is estimated at 1.4–1.6× at fixed base size on `f2m21` (desk estimate, no record of it being tried).
- **LEAD-05.** Solovyev's "degree of regularity" is the degree-capped closure with iterated falls (Huang–Kosters–Yeo Def. 1), which matches the repository's MutantXL closure and msolve's F4 step degree draw for draw and sits below XL refutation. The M4GB fix is PR #14 of `cr-marcstevens/m4gb`, opened by the preprint's author on 18 Sep 2026 and merged 25 Sep 2026. The preprint's "5 for $37 \le n \le 52$" hides a parity split (odd `n = 51` predicted 6). The decisive cells beyond the cheap test (S_4 at `N = 27`, S_3 at $N \approx 52$, chain $m = 4$, $\ell = 6$) need 37–128 GB hosts.
- **LEAD-06.** The prime-order subgroup lies in `T_n` iff `gcd(ℓ, #E(F_q)) = 1`, automatic for $\ell > q + 1 + 2\sqrt{q}$ but capped at $\ell \le$ `#T_n ≈ q^{n-1}`. The one positive is a cheaper per-residual solve (2 unknowns, 6 solutions, yield 1/2 against 3 unknowns, 64 solutions, yield 1/6 for S_4), which comes from the smaller group and does not transfer to generic curves. The card's pointer "ePrint 2014/158" is the point-compression paper; the optimal-representation paper is arXiv 1405.2733 / ePrint 2014/333.
- **LEAD-07.** The card's "(ℓ, m) = (4, 6)" is read as $m = 4$, $\ell = 6$, the cell the repository names as blocked and unmeasured; the literal $\ell = 4$, $m = 6$ reading has no repository measurement and its arithmetic is in the dossier. `crossbred_20260920/README.md` and the MutantXL README do not exist; `oracle_ladder.txt`, `e2e_panel.txt` and the MutantXL RESULTS were used instead. The published version of ePrint 2024/992 was read at source (introduction, §2, Algorithm 1, Definition 7, Theorem 10 and the examples; not the proofs or experiments); 2017/372 and 2026/360 are abstract-only. Not read: the exporter's variable order, so whether the repository's `k` enumerates x-bits or y-bits is unknown.
- **LEAD-08.** HATSolver predicts the lex Gröbner basis directly rather than learning an ordering; the relevant learned-order paper is Bunch et al., arXiv 2602.02972. The instance-level degree is a cell-level constant, identical across targets in the committed data, so there is nothing per-instance to learn; the test is replaced by a held-out-cell split with a permutation null, runnable from ml-cryptanalysis.
- **LEAD-09.** Implicit variants (never writing the resultant out) are not excluded, but they are not new either: they reduce to the bilinear descended system that LEAD-01, 05 and 07 already study. For $m = 3$ the method sits at `θ(2^n)`, the enumeration line. Euler–Petit is cited only through the repository's knowledge notes, which the dossier says.
- **LEAD-10.** Two side findings: `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §4.6 puts the (e_1, e_2) reach at 1/9, which W10's own e-encoding rows already exceed (the family value is 1/6); W10's $m = 3$ e-encoding rows were run in a box that binds, so 0.018176 and 0.024903 are lower bounds (a non-binding desk LP gives 0.0201 and 0.0269, unconfirmed in W10's Rust engine). The symmetric encoding also enlarges the root set: about `4p^{1/2}` slopes with small s_1 per genuine decomposition at `B = p^{1/2}`.
- **LEAD-11.** Corrections for `KN-LIT-6966ea` are listed in the dossier's §8; the "McGuire–Mueller trinomial bound" cited in `research/ic_next10_20261014/08-quasi-subfield/README.md:53` is their 2019 arXiv paper (1905.11755, read at source), not 2017/1262.
- **LEAD-12.** The $S \cdot T^2 \gtrsim N/|\mathrm{Aut}|$ form used across the repository is in none of the four bound papers; unique factorisation fails for elliptic-curve sums, so every elliptic-curve use of δ is the repository's derivation. The most uncertain classification is LEAD-03, with 01, 07 and 14 riding on it: a constant-factor drop in the slope of `D(n)` would move the exponent (desk check:

cost exponent per bit of n 0.81 at slope 1/20 and 0.48 at 1/40, against rho's 0.5), which is why LEAD-05's fit of D(n) is the first test.

- **LEAD-13.** Faugère–Wallet is not cited in the repository's SAT/SMT/Nagao survey. Open items: the paper's statement for L = g, where its g − L + 1 formula and the dossier's recount differ by at most 2× in degree, and whether the HCDLP step of ghs_descent.rs (the only repository code that builds a characteristic-2 cover) uses Nagao. Constant-factor levers named instead: a sieved or batched descent and cheaper line enumeration.
- **LEAD-14.** The card's premise is wrong at source: Petit–Kosters–Messeng use a composition tower, not binary digits, and their experiments are m = 2 only. Their Tables 4–5 (Magma, m = 2) independently reproduce the repository's D = 4, 5, 6 rises, which RESEARCH_PKM_TOWER_ORACLE.md §10.10 had recorded as unchecked. Kudo et al. is located as CANS 2018, LNCS 11124.

4. Ranking and what to do next

All 14 dossiers are in; the ranking uses their verdicts and LEAD-12's gate. No card produced a measured improvement, and none was expected to: this was a search round. The order below is by what a measurement would decide per unit of cost.

1. **Run LEAD-05's cheap test first** (about 1 person-day, 10–15 CPU-hours, ≤ 12 GB): replicate the S_3 closure boundaries at N = 16/17 and 36/37 with the repository's closure driver and msolve, with n = 35 as the decisive cell against Kousidis–Wiemers' Magma reading, and check three closure-dimension formula points. It decides whether the 2026 law can be imported at all, and LEAD-12 makes it the gate for 01, 03, 07 and 14.
2. **LEAD-01's residual probe R1** (1–1.5 person-days, < 4 CPU-hours): export the closed-form carry seeds and test whether msolve finishes at degree 3 on the (15, 5) and (11, 4) cells. About a 10 % prior; the prize is one degree, worth 7–65× in Macaulay width.
3. **LEAD-04's arm C** (1 person-day): the cofactor-projected base on the random-binary ladder, 1.4–1.6× estimated at fixed base size, with the prime-order negative control that must read 1.000×. A measured improvement over a leaderboard row would count under the success criterion even at that size.
4. **Do not spend on** LEAD-02, 03, 06, 09, 10, 13, 14 (closed at the desk, with the reasoning in the dossiers) or LEAD-08 beyond a cheap null test (the theorem leaves only matrix cost, where hand rules already exist). LEAD-07 is worth its 3–5 person-days only if the (4, 6) cell is wanted for LEAD-05's degree question and no 128 GB host is available; its own kill is expected to fire.
5. **Reading debt** (0.5 person-day each, needs a host outside the Cloudflare block): ePrint 2017/1262 (LEAD-11) and the elliptic-curve section of ePrint 2026/384 (LEAD-12). Until then both verdicts stay *unverifiable-from-here*.

The gap to matched rho is unchanged by this round: no row moves, and the distance remains what docs/ic/LEADERBOARD.md records.

5. Verification (mechanical tier)

Each leads/LEAD-NN.verify.md re-fetched every citation, tested every path and grepped every quoted number, recording outcomes verbatim. The non-clean rows, read by hand:

- **Citation mismatches.** All are year-of-version, wording or search artefacts: arXiv submission year against journal year (LEAD-06 ×3, LEAD-03 ×1), a search that ranked a different paper first and the right one second (LEAD-12 ×1, LEAD-09 ×1 for Ore 1933), title capitalisation, a paraphrased or abbreviated title in the dossier (LEAD-01 ×2, LEAD-07 ×1, LEAD-09 ×1, LEAD-11 ×2), a documentation page that carries no author or year (LEAD-07 ×1), two author-page PDFs that the fetch tool returns as binary (LEAD-02 ×2; the dossier read them by text extraction), and a thesis whose host (edoc-vmtest.ub.unibas.ch) did not resolve (LEAD-06). No row shows a different paper or different authors.
- **Unreachable citations.** ePrint PDFs and some abstract pages (403 or a Cloudflare challenge), one Springer chapter behind a login, the Caen 2018 server (504), Semantic Scholar's rate limit. The dossiers mark these abstract-only or recalled.
- **Missing paths.** Files the discriminating tests propose to create (examples/lead01_carry_seeds.rs, examples/lead14_block_bounds.rs, examples/pdp_perm_f4.rs, csd/order_probe.py), one external repository file (cr-marcstevens/m4gb src/m4gb.hpp), and one brace-expansion artefact (README for README.md). No cited existing path is missing.
- **Numbers not found by literal grep.** Formatting differences ("0 of 345" for "0/345", "8 of 8", "N = 40 and 42", superscripts such as 2^{150} for 2^150 in LEAD-03; confirmed by hand in the named files), and in LEAD-14 three numbers that live in research/pkm_m3_n21_msolve_20261018/RESULTS.md while the dossier names the sibling README.md (an attribution slip, not a wrong number). One LEAD-05 number ("37 of 38" in fall_degree_bounds_20261008/RESULTS.md) was not located by hand either and stays unconfirmed.

6. Corrections to repository records surfaced by this round

Recorded here for the owners of those files; this PR changes none of them.

- knowledge/literature/KN-LIT-6966ea.md (autoresearcher) is abstract-only and should carry the corrections in leads/LEAD-11.md §8.

- `research/ic_next10_20261014/08-quasi-subfield/README.md:53` cites a "McGuire–Mueller trinomial bound" to ePrint 2017/1262; the source is arXiv 1905.11755.
- `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §4.6's `(e₁, e₂)` reach of 1/9 is a basic-strategy figure; the family value is 1/6 (LEAD-10).
- `RESEARCH_PKM_TOWER_ORACLE.md` §10.10's "unchecked" against Petit–Kosters–Messeng: their Tables 4–5 reproduce the `D = 4, 5, 6` rises (LEAD-14).
- `LEADS.md` card errors found by the dossiers and left in place for the record: the Solovyev preprint is 2026, not 2025 (corrected in the card heading and figures); "ePrint 2014/158" is the point-compression paper (LEAD-06); HATSolver does not learn orderings (LEAD-08); PKM use a composition tower, not binary digits (LEAD-14); "`|A|` in every oracle" overstates the pair-table saving (LEAD-04).

7. Unresolved checks and access failures

- IACR ePrint PDFs are blocked from this container by a Cloudflare challenge; WebFetch on the same URLs returns 403. The texts that matter most and were not read: ePrint 2017/1262 (LEAD-11), the elliptic-curve section of ePrint 2026/384 (LEAD-12), ePrint 2015/573 (LEAD-05), ePrint 2019/313 (LEAD-08; covered by Trimoska's thesis, read at source).
- The Caen 2018 talk and paper by Mueller (nitaj.users.lmno.cnrs.fr) returned 504; Mueller's UCD thesis was not found online.
- Two LEAD-07 agents produced nothing (one lost to the container restart, one stalled and stopped after three hours); the third wrote the dossier under a 25-minute budget, so its reading of the Crossbred literature is shallower than the other dossiers' and it says so.
- The published version of ePrint 2024/992 and the two Joux–Vitse PDFs were read from saved copies by text extraction (the fetch tool returns binary for them); Faugère–Wallet's HAL and Springer pages were unreachable, so LEAD-13 rests on the abstract and Wallet's 2016 slides.

8. Graphs checked and left unchanged

No measured result was produced, so no canonical graph changes: `docs/index-calculus-scoreboard.html`, `docs/ic/progress-timeline.json`, `docs/ic-leaderboard.html` / `docs/ic/LEADERBOARD.md` / `docs/ic/leaderboard.json`, `docs/performance-gains/`, and `benchmarks/README.md` are untouched. Nothing here is a performance result, so no `benchmarks/` entry is owed.

10. Round 2 (2026-10-23): five leads from the dossiers' own pointers

Cards in `LEADS.md`, "Round 2 cards"; dossiers and verification cards in `leads/`. Same briefs and rules as round 1. Verification totals are the mechanical card's (citations verified / mismatch / unreachable; paths exist / missing); every non-clean row was read by hand and is classified in §10.2.

Lead	Card (class)	Dossier verdict
15	Plane-quartic Jacobian arithmetic for Diem's Õ(q) branch of the cubic-GHS transfer (transfer, exponent)	**exponent-candidate**, with the card mis-aimed: Diem–Thomé's practical algorithm uses about 12 class-group additions and needs no unique class forms (Riemann–Roch gives Diem's divisor "reduced along P_0" where one is needed), and the line harvest uses no Jacobian arithmetic at all, so the missing arithmetic is a small cost term, not the blocker; Diem's JTNB 2012 paper proves Õ(q) for plane quartics (the census README calls it heuristic), but that proved version needs about $48q$ additions and is not competitive while the practical one is; the Flon–Oyono–Ritzenthaler group law costs 148M + 15S + 2I per addition (about 181 multiplication-equivalents at $q = 257$), inside the card's 10^3 kill line; the untested links are the line yield on the bielliptic GHS quartics and whether `C(F_q)` reaches the ℓ-part at $q \geq 257$
16	Nagao's quadratic system against the D_4-symmetrised S_5 at $k = 4$ (formulation)	**closed-by-theorem**: Nagao's 12 quadrics in 12 unknowns define the same 4096-point variety as Gaudry's S_4 form in other coordinates; only the central 2-torsion translation acts on it, so its best quotient is 2048 against 512 for the D_4 system, and the card's second kill fires; the predicted cost is $3 \cdot 10^{10}$–10^{12} multiplication-equivalents per decomposition against the committed D_4 solve; the card's "about 85 solutions" is $4096/48$, a cost ratio read as a count (the repository's quotient is 512); the better-shaped Nagao system at $k = 4$ is on the genus-2 cover of Joux–Vitse §5 (4 quadrics in 4 unknowns, degree 16, read at source), which has no card and no repository route
17	Closed-form closure dimensions for the descended S_3 chain (theory)	**unverifiable-from-here** for the degree law, with the count closing exactly where it can be checked: closed forms for the degree-2 and degree-3 closure dimensions and for the plain degree-4 Macaulay rank, for $m = 3, 4$ and $\ell \geq 4$, reproduce the committed $(m, n, \ell) = (3, 11, 5)$ cell exactly ($XL_3 = 307$, $V_3 = 772$, $XL_4 = 4071$), are exact for XL_3 and XL_4 in 16 of 17 committed cells with $\ell \geq 4$ and within ±2 for V_3 in 16, and predicted the $m = 4$, $\ell = 6$ cells at $n = 9, 10$ exactly in desk runs; the obstruction is that the quadric-fall space is not stable from $D =$

		3 to D = 4 (it grows to about 1.5–2.2× the degree-3 value 3n + 2ℓ on the unsaturated m = 3 cells), so the degree-3 count does not fix the saturation degree and no prediction is made at (4, 6); the 4ℓ lag term and the (m − 1)(n + ℓ) degree-4 term are fitted, not derived
18	Structured elimination for C_4: sparse FGLM or resultants on the D_4 quotient (solver)	**closed-by-measurement**: the card's premise is stale; the ideal degree is 512, not 85, and the repository's chains step (landed 2026-10-14) already does what the card proposes (only the dense columns of the multiplication matrix, as in Faugère–Mou and msolve §4.4), taking C_4 from $2.54 \cdot 10^{10}$ to $6.9–8.2 \cdot 10^{7}$; the remaining paths are at parity (the det A(λ) step costs $2.5 \cdot 10^{7}$ against $2.3 \cdot 10^{7}$ for the Krylov step), a resultant chain costs about $3 \cdot 10^{10}$ (dense desk bound, roughly 370× the current C_4), root finding is 0–4 % of C_4, and msolve has no weighted order and cannot replay its F4 trace across targets
19	Cofactor-projected factor base on the random-binary ladder (filter)	**constant-factor candidate** on the generic line: the raw subspace base is never projected on the random-binary ladder (run_char2_instance), while the lever already exists for Koblitz bases (build_subgroup_orbit_factor_base); about 50 % of the folded pair table lies in the subgroup on f2m15 and f2m21 and 16.7 % on f2m27, walked targets always stay in the subgroup; the published rows also carry a class-test charge (about 12 of 26.3 units on f2m15) that the projected arm never pays, which is accounting, not an improvement, so the protocol measures against the raw base with that charge removed and reports both ratios

Table D. Ranking inputs and verification totals, round 2.

Lead	Confidence	Expected gain if it works	Test cost	Verification (citations v/m/u; paths e/m)
15	medium (Diem–Thomé ran the same harvest up to q = 2^31; the toy quartic IC passes 90/90; yield on GHS quartics and the ℓ-part reach are untested)	desk estimate: cost ratio to the Item-7 rho model about 1.1 at q = 257, 0.25 at 2^11, 0.085 at 2^15 (3.3 / 1.1 / 0.40 with linear-algebra multiplications mod ℓ priced at 9 field multiplications); fitted exponent about 1.0 against 1.30 for the hyperelliptic branch and 1.5 for rho; would move the m = 3 double-large-prime row at q = 2^15 from $2.60 \cdot 10^{8}$ to about $1.4 \cdot 10^{7}$, but only on instances with a plane-quartic cover (a new row)	6–10 person-days, < 1 CPU-h; the harvest alone about 3 days, mostly an F_{q³} quartic builder at q ≥ 257 (none exists above q = 11)	6 / 5 / 4; 13 / 0
16	high (symmetry and degree derived and desk-checked at p = 13); medium (cost transferred from published timings)	none; the card's C_4 of $2.54 \cdot 10^{10}$ is also stale (now $6.9 \cdot 10^{7}$), so Nagao is predicted 450–14,000× worse than the current baseline	1–1.5 person-days of Rust (no Nagao exporter exists), < 3 CPU-h, < 12 GB; not recommended	2 / 1 / 7; 18 / 0
17	medium (degree ≤ 3 formulas); low (any D law)	none as an improvement, a pricing input only; the m = 4 degree-4 row stays at +7.62 bits to rho	replaying the formulas on the committed records: < 1 CPU-minute; the new m = 4, n = 10, ℓ = 6 cells about 40 CPU-minutes at 8–10 GB; degree 5 needs a 70–128 GB host	4 / 1 / 0; 18 / 0
18	medium (dimension and phase shares measured; msolve cost unmeasured, msolve not on PATH)	1.0–1.4× on the current C_4 of $8.2 \cdot 10^{7}$ at p = 1009, 1.9× at best (about 3.6 bits on the crossover); distance to rho unchanged ($1.39 \cdot 10^{6}$× rho-negation at p = 29)	0.5–1 person-day, about 40 CPU-minutes, no new Rust (msolve -g 1, -g 2, -P 1, one thread, on four plain D_4 instances against the repository's four solver arms)	7 / 0 / 5; 17 / 0
19	medium (mechanism checked exactly on the three curves; effect on S is a desk model over 3-seed means with heavy tails)	lever only: about 1.4–1.5× on f2m15 and f2m21, about 3× on f2m27; against the published rows about 2.8×, 2.1×, 3.4×; distance to matched rho from 13.1×, 18×, 38.2× to about 4.7×, 8.7×, 11×, still above rho on every row	about 1 person-day of Rust (150–200 lines with tests, a --base-arms flag on ic boundary), < 5 CPU-minutes, < 0.5 GiB; landing a row adds the validation gate	3 / 0 / 3 (+3 unverified); 18 / 2

10.1 What the round-2 dossiers change in the ranking

1. **LEAD-19 moves to the top of the measurable list.** It is the only item with a desk estimate of an improvement over existing leaderboard rows (random-binary `f2m15`, `f2m21`, `f2m27`), it needs about a person-day of Rust and minutes of CPU, and its protocol carries the prime-order negative control (`fp24`, must read 1.000×) and the accounting separation (the class-test charge) that the success criterion demands. Expected result if it works: three rows improved by 1.4–3×, still above rho.
2. **LEAD-15 is the only exponent lead left on the board**, and it is a transfer: Diem's practical line-intersection index calculus on the plane-quartic branch of the cubic-GHS cover, with a desk cost ratio to rho that falls from about 1.1 at `q = 257` to 0.085 at `q = 2^15`. The work is an `F_{q³}` instance builder for plane-quartic covers at `q ≥ 257` and the line harvest (about 3 person-days before any Jacobian arithmetic is needed), on the weak class only.
3. **LEAD-18 and LEAD-16 retire the C_4 engineering angle**: the sparse-FGLM idea is already in the tree, and Nagao's coordinates cannot take the D_4 quotient. The pointer worth a card is Nagao on the genus-2 cover at `k = 4` (Joux–Vitse §5).
4. **LEAD-17 does not replace the measurement.** The combinatorial count is exact through degree 3 and for the plain degree-4 rank, which is a useful zero-compute check on every committed cell, but the quadric-fall space grows between degrees 3 and 4 by a factor the count does not predict, so the saturation degree at `(4, 6)` still needs the run (about 40 CPU-minutes at 8–10 GB for degree 4; a 70–128 GB host for degree 5). The dossier reads the card's "(ℓ, m) = (4, 6)" as `m = 4`, `ℓ = 6`.

10.2 Verification notes, round 2

- **LEAD-15.** Five "mismatch" rows are PDFs the fetch tool returns as binary (the HAL copy of Diem–Thomé, the JTNB paper, the Flon–Oyono–Ritzenthaler report), a 502 from the ECC 2005 slides mirror, and a Springer page behind a login; none shows a different paper. The dossier read Diem–Thomé and the JTNB paper by text extraction.
- **LEAD-16.** The one mismatch is a title the dossier itself marks "seen, not fetched". Six unfound numbers are rounded forms.
- **LEAD-18.** Twenty unfound numbers are rounded scientific notation (`2.54·10¹⁰` for the files' `2.535–2.552·10¹⁰` and full-digit JSON values).
- **LEAD-17.** One genuine defect: the arXiv identifier the dossier guesses for Caminata–Gorla (solving degree versus degree of regularity) resolves to an unrelated paper, so that citation stands as recalled, not verified; one unfound number; all 18 paths exist.
- **LEAD-19.** No mismatches; three citations the card could not route (no identifier) are listed as unverified; the two missing paths are the output files the proposed test would write (`research/lead19_projected_base/{char2,prime24}.json`).

9. Files

File	What it is
`LEADS.md`	The 14 lead cards and the standing constraints
`BRIEF_SONNET.md`, `BRIEF_HAIKU.md`	The director-tier and mechanical-tier briefs
`leads/LEAD-NN.md`	Director-tier dossiers (Sonnet 5.5), one per card
`leads/LEAD-NN.verify.md`	Mechanical-tier verification cards (Haiku 5.5), one per dossier
`papers.csv`	Literature map, 44 sources, access status per source
`figures/pdp_landscape.svg`, `.png`, `diagram.py`	Figure 1 and its source
`figures/dreg_records.svg`, `.png`, `.csv`, `plot_dreg.py`	Figure 2, its data and its source
`report.typ`, `report.pdf`, `build.sh`	This report rendered (pandoc → Typst), and the build script